

Decision Tree Customer Satisfaction Analysis

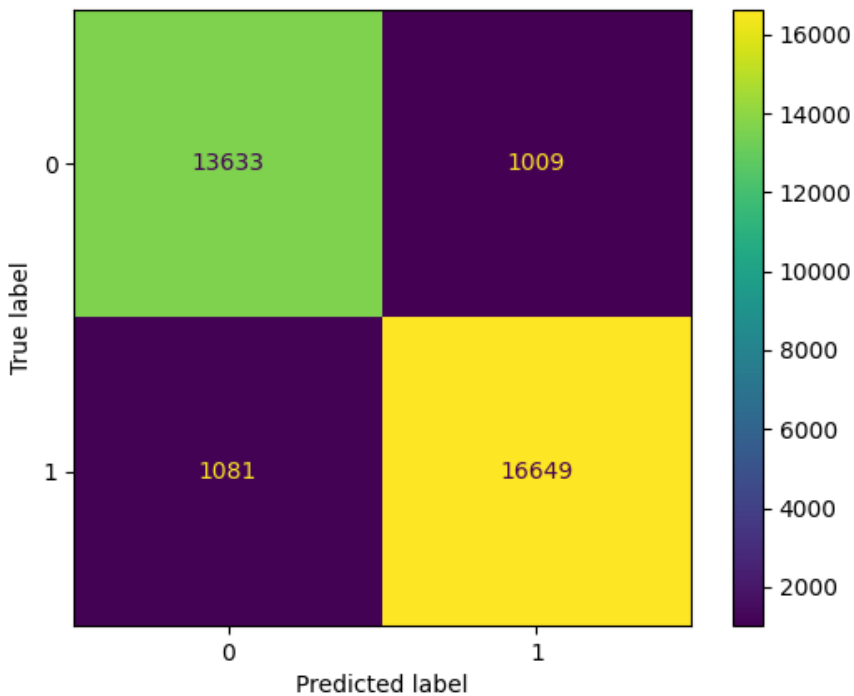
Model Performance:

Accuracy: 0.935
Precision: 0.943
Recall: 0.939
F1 Score: 0.941

The model demonstrates strong predictive performance, correctly classifying customer satisfaction outcomes over 94% of the time.

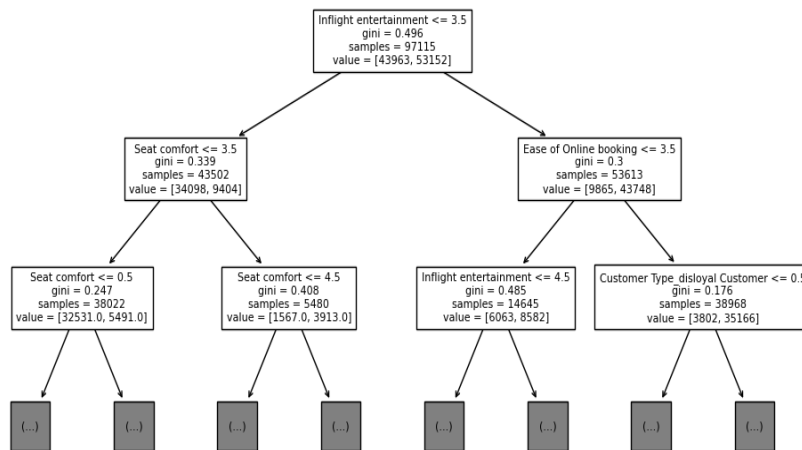
Confusion Matrix Interpretation

The confusion matrix shows a high concentration of true positives and true negatives, indicating the model reliably distinguishes between satisfied and dissatisfied customers. False positives and false negatives are relatively low, suggesting minimal misclassification risk.



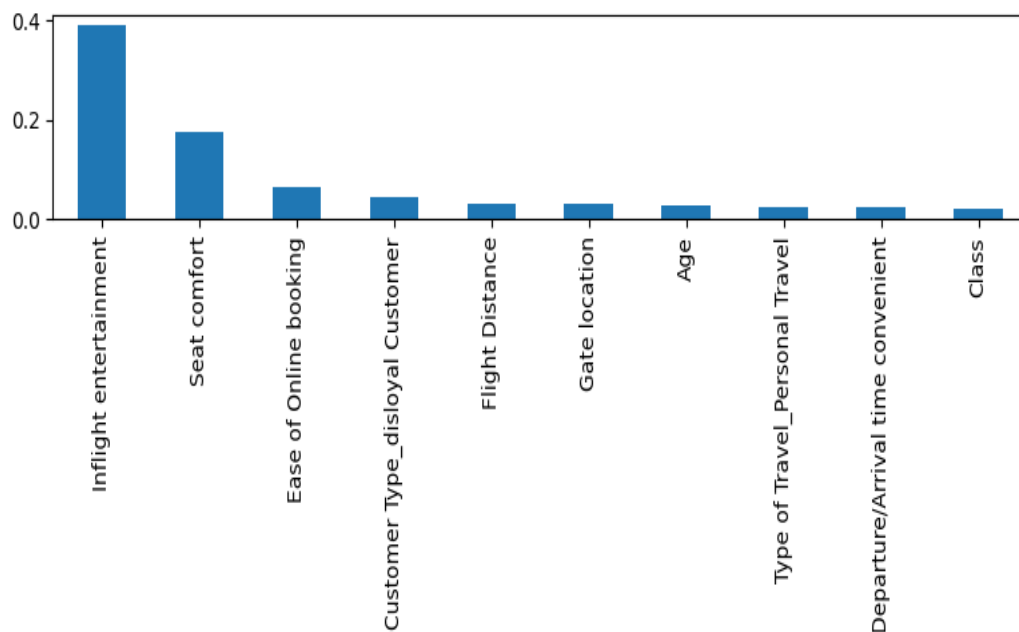
Decision Tree Structure

The top splits in the decision tree highlight the most influential factors driving customer satisfaction. Early splits are dominated by service-quality variables, showing these attributes are decisive in customer perceptions.



Feature Importance Insights

Feature importance analysis indicates that inflight entertainment, seat comfort, and ease of online booking are the strongest predictors of satisfaction. These variables account for the majority of impurity reduction in the model.



Actionable Business Recommendations

- Prioritize investment in inflight entertainment systems and seat comfort upgrades.
- Streamline and improve the online booking experience to reduce friction.
- Use this model operationally to proactively identify at-risk customers and intervene before dissatisfaction leads to churn.